

End-to-End Process · Order-to-Cash (O2C)

PCF 3.5 Manage Sales Orders · PCF 4.4 Manage Logistics and Warehousing · PCF 9.4 Customer Credit, Invoicing & AR

View 1 — HITL Design

O2C HITL Design Summary

O2C Aggregate H% · M3	Decision Gates	Transaction Steps	Exception Flows	Δ M1 → M3
52% Human	4 · 80% H	12 · 35% H	4 · 82% H	-16%

H% by Pattern Type at Each Maturity Level

Pattern	Nature of Work	M1 Undesigned	M2 Emerging	M3 ★ Designed	M4 Optimized	M5 Leading
Decision	Judgment & Authority	92%	88%	80%	72%	65%
Knowledge	Synthesis & Interpretation	85%	78%	65%	52%	42%
Document	Content Generation	72%	58%	40%	28%	18%
Transaction	Rules-Based Processing	60%	48%	35%	22%	12%
Exception	Non-Standard Resolution	95%	92%	82%	75%	68%

Pattern Type Legend

Decision 80% H at M3	Knowledge 65% H	Document 40% H	Transaction 35% H	Exception 82% H
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O2C End-to-End Process — HITL Design · PCF-Derived Process Ownership

Swim lanes = PCF process group ownership · PCF L3 reference shown on each step · M3 H% = HITL design intent

Sales PCF 3.5 PCF 3.5	Materials PCF 4.4 PCF 4.4	Finance PCF 9.4 PCF 9.4	Customer External External participant
Decision gate 3.5.4 Accept & Validate Sales Order DECISION 80%	4.4.3 Pick / Pack / Ship TRANSACTION 35%	9.4.2 Generate Invoice DOCUMENT 40%	Ext Submit Sales Order DOCUMENT 40%
3.5.4 Determine Availability TRANSACTION 35%	4.4.3 Track Inventory Accuracy TRANSACTION 35%	9.4.2 Apply Taxes TRANSACTION 35%	Ext → Sales (3.5.4)
Decision gate 3.5.4 Determine Fulfillment Process DECISION 80%	4.4.4 Plan & Execute Outbound Delivery TRANSACTION 35%	9.4.2 Distribute Invoice TRANSACTION 35%	Ext Receive Order Confirmation TRANSACTION 35%
3.5.4 Confirm Sales Order DOCUMENT 40%	4.4.4 Track Carrier Delivery Performance TRANSACTION 35%	9.4.2 → Customer + AR (9.4.3)	Ext Receive Goods / Delivery TRANSACTION 35%

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3.5.4 → Materials (4.4.3) + Customer	4.4.4 → Finance (9.4.2) + Customer	9.4.3 Maintain AR TRANSACTION 35%	Ext Receive Invoice TRANSACTION 35%
		9.4.3 Monitor Aging TRANSACTION 35%	Ext Submit Payment TRANSACTION 35%
		9.4.3 Reconcile Accounts TRANSACTION 35%	Ext → Finance (9.4.5)
		9.4.5 Process Customer Payment TRANSACTION 35%	Ext Receive Payment Confirmation TRANSACTION 35%
		9.4.5 Apply Cash TRANSACTION 35%	
		9.4.5 → Customer (Payment Confirmation)	

Exception Flows — Human Resolution Required (82% H at M3)

3.5.4 Process Backorders & Updates EXCEPTION 82%		9.4.4 Manage Collections / Dunning EXCEPTION 82%	
3.5.4 Handle Order Inquiries EXCEPTION 82%		9.4.4 Resolve Disputes EXCEPTION 82%	

HITL Design Insight

HITL Design Insight — O2C

4 Decision gates (80% H at M3) are the irreducibly human layer — order acceptance, availability determination, and fulfillment routing carry commercial and customer authority that must remain human by design. **12 Transaction-pattern steps** (35% H at M3) represent the AI absorption opportunity — fulfillment, invoicing, AR maintenance, payment processing, and cash application in rules-based sequence. **4 Exception flows** (82% H) require human resolution — backorders, order inquiries, collections/dunning, and dispute resolution. O2C aggregate at M3: **52% Human / 48% AI** — grounded in APQC PCF SIPOC reference architecture.

HITL Impact Analysis

H% = Human involvement at each maturity level · M3 = HITL Design Intent

Process Step (L3)	PCF	Pattern	M1 H%	M3 H% ★	Δ M1→M3	AI Opportunity
O2C End-to-End Process · L1 Aggregate	68%		68%	52%	-16%	Significant
PCF 3.5 · Manage Sales Orders						

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PCF 3.5 Manage Sales Orders · PCF 4.4 Manage Logistics and Warehousing · PCF 9.4 Customer Credit, Invoicing & AR

↳ Accept & Validate Sales Order	3.5.4	Decision	92%	80%	-12%	Retain Human
↳ Determine Availability	3.5.4	Transaction	60%	35%	-25%	High
↳ Determine Fulfillment Process	3.5.4	Decision	92%	80%	-12%	Retain Human
↳ Confirm Sales Order	3.5.4	Document	72%	40%	-32%	High
↳ Process Backorders & Updates	3.5.4	Exception	95%	82%	-13%	Retain Human
↳ Handle Order Inquiries	3.5.4	Exception	95%	82%	-13%	Retain Human
PCF 4.4 · Manage Logistics and Warehousing						
↳ Pick / Pack / Ship	4.4.3	Transaction	60%	35%	-25%	High
↳ Track Inventory Accuracy	4.4.3	Transaction	60%	35%	-25%	High
↳ Plan & Execute Outbound Delivery	4.4.4	Transaction	60%	35%	-25%	High
↳ Track Carrier Delivery Performance	4.4.4	Transaction	60%	35%	-25%	High
PCF 9.4 · Manage Customer Credit, Invoicing, and Accounts Receivable						
↳ Generate Invoice	9.4.2	Document	72%	40%	-32%	High
↳ Apply Taxes	9.4.2	Transaction	60%	35%	-25%	High
↳ Distribute Invoice	9.4.2	Transaction	60%	35%	-25%	High
↳ Maintain AR	9.4.3	Transaction	60%	35%	-25%	High
↳ Monitor Aging	9.4.3	Transaction	60%	35%	-25%	High
↳ Reconcile Accounts	9.4.3	Transaction	60%	35%	-25%	High
↳ Manage Collections / Dunning	9.4.4	Exception	95%	82%	-13%	Retain Human
↳ Resolve Disputes	9.4.4	Exception	95%	82%	-13%	Retain Human
↳ Process Customer Payment	9.4.5	Transaction	60%	35%	-25%	High
↳ Apply Cash	9.4.5	Transaction	60%	35%	-25%	High
External Participant · Customer						
↳ Submit Sales Order	Ext	Document	72%	40%	-32%	High
↳ Receive Order Confirmation	Ext	Transaction	60%	35%	-25%	High
↳ Receive Goods / Delivery	Ext	Transaction	60%	35%	-25%	High
↳ Receive Invoice	Ext	Transaction	60%	35%	-25%	High
↳ Submit Payment	Ext	Transaction	60%	35%	-25%	High
↳ Receive Payment Confirmation	Ext	Transaction	60%	35%	-25%	High

AI Capability Map — O2C

PCF 3.5 Sales Order Management	PCF 4.4 Logistics & Warehousing	PCF 9.4 Invoicing & Accounts Receivable	External Customer
• Order capture via EDI / portal / AI chat	• AI-directed pick path optimization	• Auto-invoice generation from billing trigger	• Self-service order portal
• Credit & availability auto-validation	• Inventory reconciliation & variance detection	• Tax calculation — Vertex / Avalara	• Real-time order tracking

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• Fulfillment routing optimization	• Carrier selection & route optimization	• Invoice distribution via EDI / portal	• AI-assisted inquiry resolution
• Order confirmation auto-generation	• SLA monitoring & exception alerting	• AR aging analysis & risk scoring	• Electronic invoice receipt (EDI)
• Backorder triage & customer communication	• ASN generation & carrier EDI	• Dunning notice automation	• Payment scheduling automation
• Inquiry resolution via AI chatbot	• Proof of delivery & billing trigger	• Cash application & remittance matching	• Payment confirmation & remittance